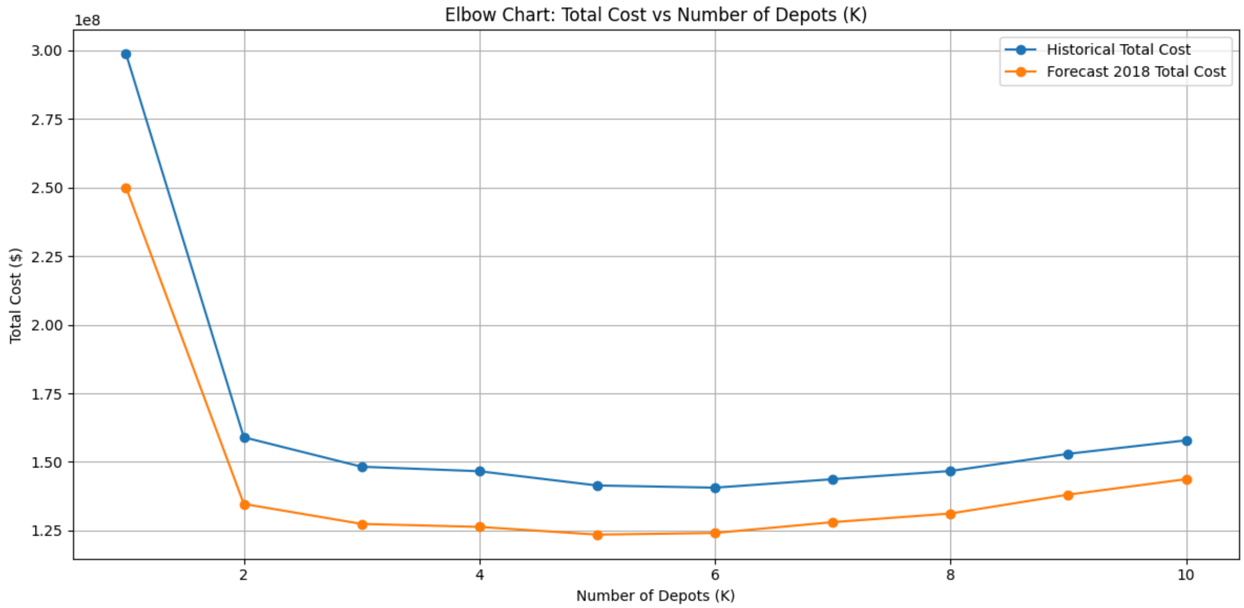
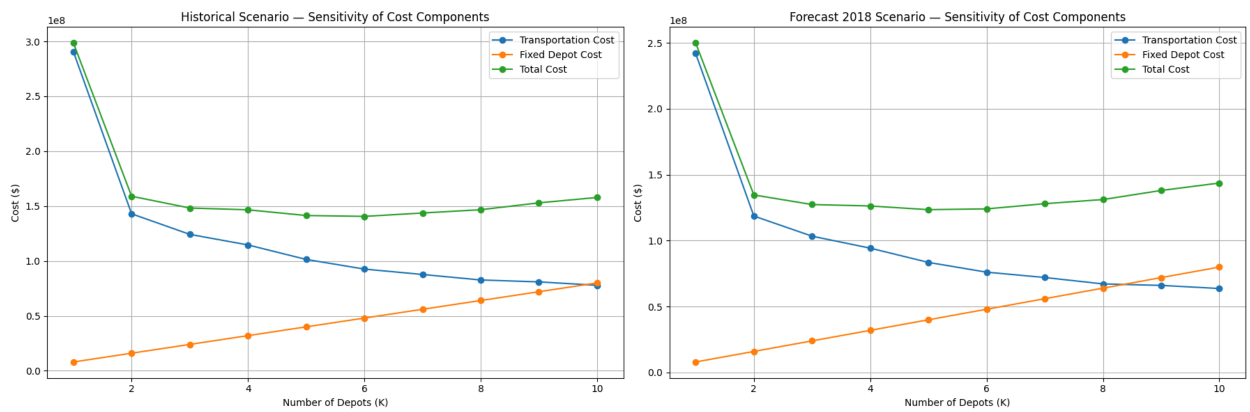
Historical K summary shape: (10, 10)
Forecast K summary shape: (10, 10)





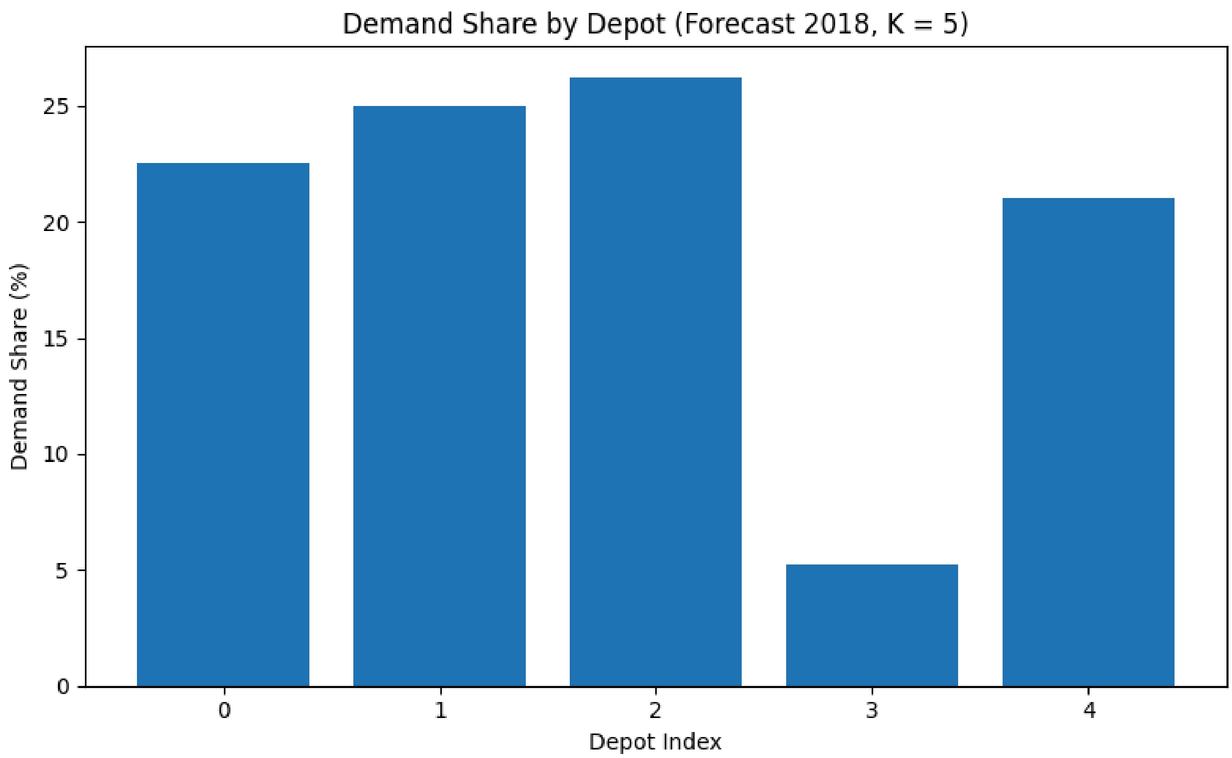
Figures saved:
Elbow_Chart_Total_Cost_vs_K.png
Sensitivity_Chart_Cost_Components.png

Final K=5 design re-computed. LA depot: 1 Houston depot: 2

Extra Insight 1 – Demand balance by depot (K = 5):

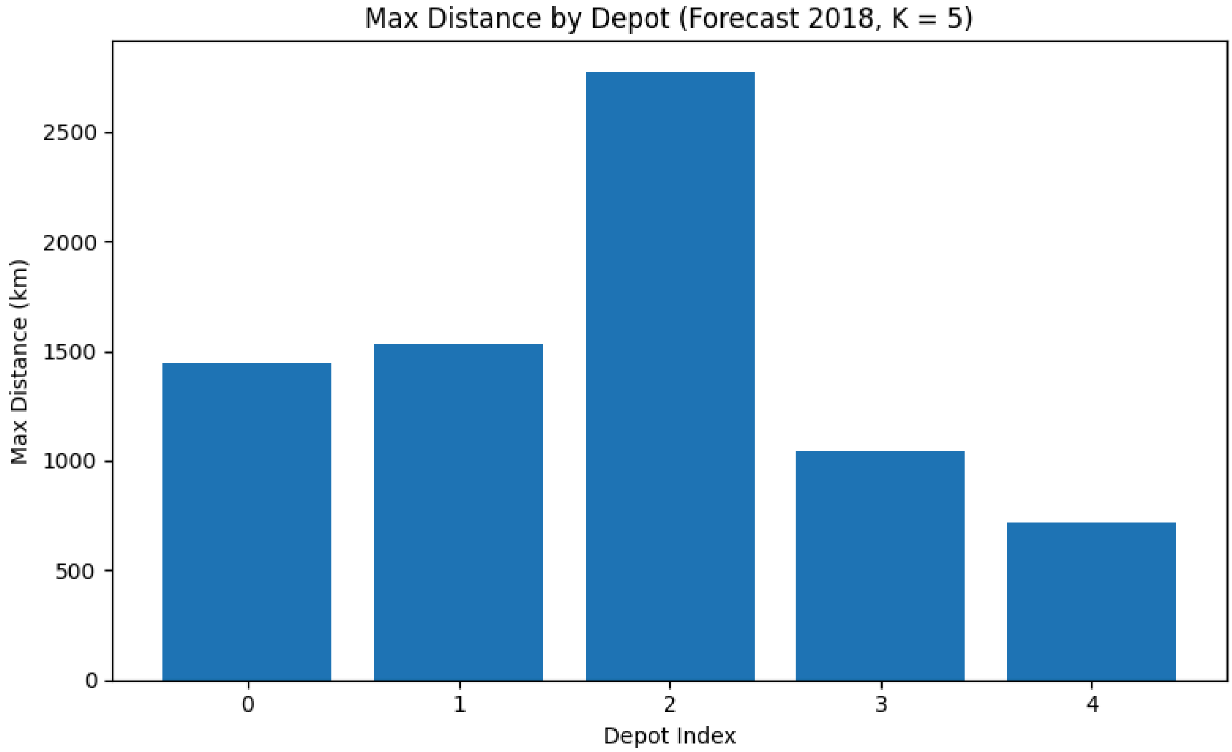
	Assigned_Depot	Total_Demand	Demand_Share_pct
0	0	35566.290252	22.538141
1	1	39437.220699	24.991126
2	2	41416.871894	26.245618
3	3	8197.564068	5.194746

4	4	33186.953086	21.030369



Extra Insight 2 – Distance fairness by depot (K = 5):

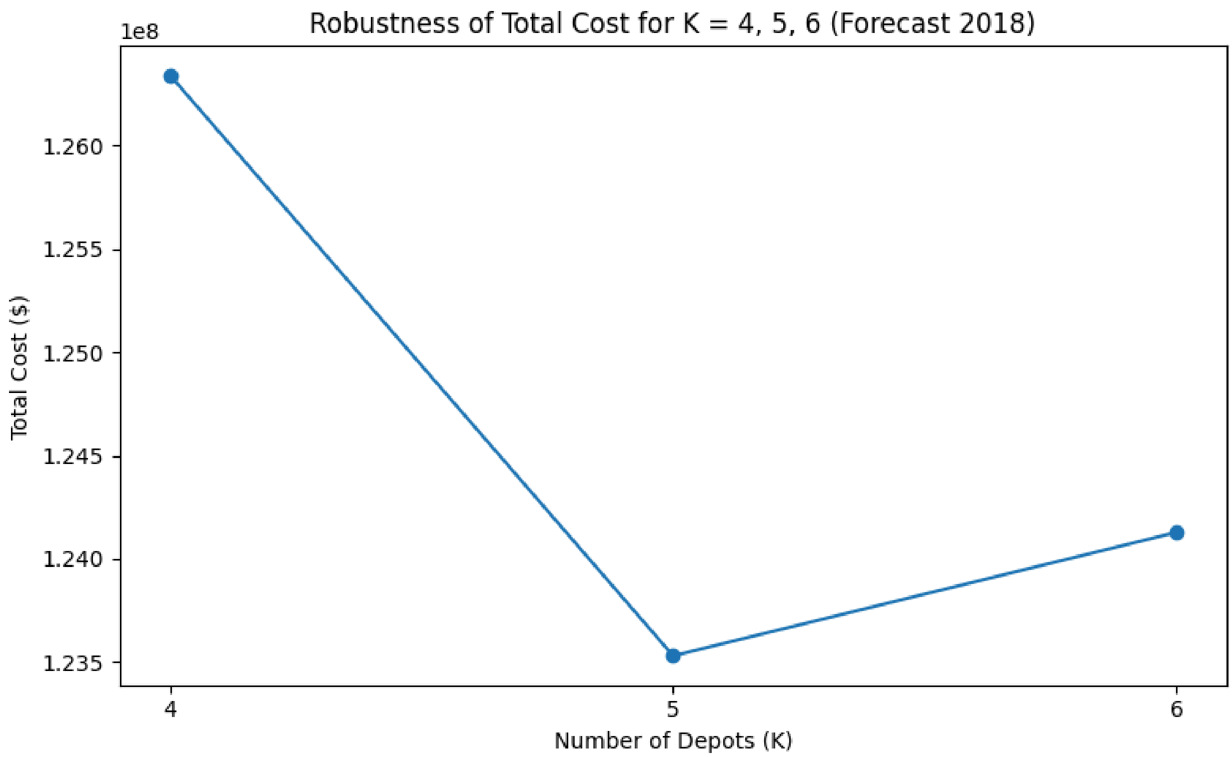
	Assigned_Depot	Avg_Distance_km	Max_Distance_km
0	0	615.401046	1443.895260
1	1	455.190165	1529.967096
2	2	1081.409723	2775.814609
3	3	310.005938	1043.154083
4	4	272.301907	721.660665




```
/tmp/ipython-input-2868685415.py:128: FutureWarning: Calling float on a single element Series is deprecated and will raise a TypeError in the future. Use float(ser.iloc[0]) instead
  k5_cost = float(robust.loc[robust["K"] == 5, "Total_Cost"])

Extra Insight 3 – Robustness comparison (K = 4, 5, 6):
   K  Transportation_Cost  Fixed_Depot_Cost    Total_Cost  \
3  4         9.434097e+07        32000000.0  1.263410e+08
4  5         8.353076e+07        40000000.0  1.235308e+08
5  6         7.612812e+07        48000000.0  1.241281e+08

   Total_Cost_pct_vs_K5
3              2.274911
4              0.000000
5              0.483572
```



Extra insight figures saved as:
ExtraInsight1_Demand_Balance_K5.png
ExtraInsight2_Fairness_MaxDistance_K5.png
ExtraInsight3_Robustness_K4_K5_K6.png